(15) State how continuity is destroyed at $x = x_0$ for each of the following graphs.

(a)

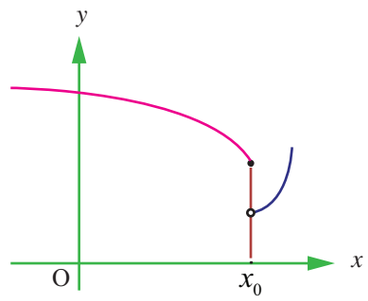


Fig. 9.38

(b)

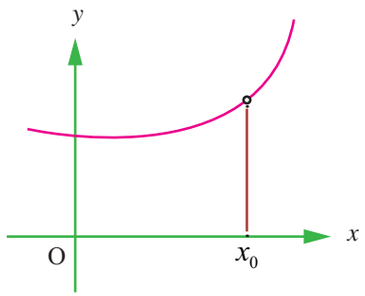


Fig. 9.39

(c)

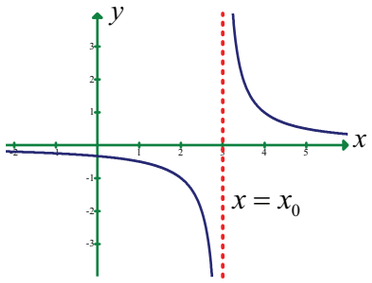


Fig. 9.40

(d)

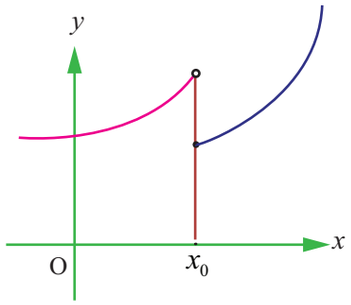


Fig. 9.41

EXERCISE 9.6

Choose the correct or the most suitable answer from the given four alternatives.

(1) $\lim\limits_{x\to\infty} \dfrac{\sin x}{x}$

(1) 1 (2) 0 (3) ∞ (4) $-\infty$

(2) $\lim\limits_{x\to\pi/2} \dfrac{2x-\pi}{\cos x}$

(1) 2 (2) 1 (3) -2 (4) 0

(3) $\lim\limits_{x\to0} \dfrac{\sqrt{1-\cos 2x}}{x}$

(1) 0 (2) 1 (3) $\sqrt{2}$ (4) does not exist

(4) $\lim\limits_{\theta\to0} \dfrac{\sin\sqrt{\theta}}{\sqrt{\sin\theta}}$

(1) 1 (2) -1 (3) 0 (4) 2

(5) $\lim\limits_{x\to\infty} \left(\dfrac{x^2+5x+3}{x^2+x+3}\right)^x$ is

(1) e^4 (2) e^2 (3) e^3 (4) 1

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